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9 February 1979
SUPERSEDING
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17 December 1954

MILITARY SPECIFICATION

CANISTER, CHEMICAL-BIOLOGICAL MASK: M10A1

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of canister for a gas mask.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel and Seals.
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.
PPP-B-621 - Boxes, Wood, Nailed and Locked-Corners.
PPP-B-636 - Box, Fiberboard.

FSC 4240

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-101 - Preservation, Packaging and Packing Materials, Test Procedures.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipments and Storage.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

E5-3-547 - Canister, Chemical-Biological, M10A1 Assembly.
B136-14-170-5 - Tester, Air Leakage, Flow Type, Q211.
B136-44-1-5 - Resistance Indicator, Inlet and Outlet Valve, Q212.
B136-41-850-3 - Test Apparatus Agent Gas Sorption Q216.
B136-42-102-11 - Penetrometer, Filter Testing, DOP, Q127.
C38-5-296 - Meter, Chemical Container, Volume Q134.
C26-19-51 - Machine, Rough Handling, Q113.
B38-6-351-1 - Indicator, Resistance, Canister, Air Purifier and Chemical Container, Q112.
B136-41-600-11 - All Purpose Apparatus, Gas Life Testing, Q95.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. All materials and components used in the manufacture and assembly of the canister shall conform to the specifications and drawings forming a part of this specification.

3.2 Manufacture and assembly. The canister shall be manufactured and assembled as specified on Drawing E5-3-547 and subsidiary drawings listed thereon.

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3.3 Chemical container leakage. The chemical container top assembly when fabricated in accordance with Dwg E5-3-552 and tested at the swaged joint shall not leak when subjected to an air pressure differential of 5 psi for 10 seconds when tested as specified in 4.4.4.1. When alternate design (Dwg B5-3-555) is employed this requirement is not required. A leak is defined as the formation and release of a single bubble within the time frame.

3.4 Filling. The chemical container shall be free fall-filled with not less than 330 milliliter of impregnated charcoal when tested as specified in 4.4.4.1. After filling, the surface of the charcoal may be level or slightly above the edge of the chemical container body, thus requiring the container bottom (Dwg C5-3-550) to be pressed into the filled container until the flange is seated against the end of the container body. During the press closing operation, the filled container, with bottom in position, shall be subjected to a uniform vibration of sufficient magnitude to cause the bottom to be seated properly in the filled container, but not so severe as to cause damage or deformation to the container parts. The bottom shall be securely attached to the container body as shown on Drawing D5-3-549. The filter material on the finished chemical container shall form a firm covering, free from cuts, wrinkles, tears, or other surface irregularities and shall be securely crimped. Care shall be taken to prevent rubbing or scuffing the filter material during the wrapping, crimping and assembly operations.

3.5 Charcoal moisture content. The average moisture content of the completed canister shall not exceed 4.0 percent by weight and no single values shall exceed 5.0 percent by weight when tested as specified in 4.4.4.1.

3.6 Inlet valve disk.

3.6.1 Valve leakage. Air leakage of the valve shall not exceed 250 milliliters a minute when tested as specified in 4.4.4.1.

3.6.2 Valve resistance. The resistance of the valve shall not be more than 5 millimeters of water when tested as specified in 4.4.4.1.

3.7 Canister resistance. The air-flow resistance of the complete canister shall not exceed 75 millimeters of water at flow rate of 85 liters per minute when tested as specified in 4.4.4.1.

3.8 Canister leakage. The assembled canister (Dwg E5-3-547) shall not leak when tested as specified in 4.4.4.1. A leak is defined as the formation and release of a single bubble within the time frame.

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3.9 Gas life. The computed gas life mean minus k (see 4.3.3 and 4.4.5) times the standard deviation (see 6.5) shall be equal to or greater than values shown in table I when tested as specified in 4.4.4.1.

TABLE I. Gas life

Gas	Rough handling	Min. life, minutes
DMMP	Before and after	200.0
GB	Before	230.0
CK	Before	40.0

3.10 DOP aerosol penetration. The canister, both before and after rough handling, shall show a DOP penetration of not more than 0.025 percent before and after rough handling at a flow rate of 32 liters per minute when tested as specified in 4.4.4.1.

3.11 Rough handling. The canister body shall not have any breaks or open seams, which will allow grit or other foreign matter into the canister, when tested for 15 minutes as specified in 4.4.4.1.

3.12 Preproduction. Prior to the start of regular production, a preproduction sample shall be produced in accordance with this specification for examination and tests.

3.13 Workmanship. The canisters shall be free from any cracks, rupture seam, defects, contamination or damage may effect the intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Government responsibility. Unless otherwise specified, the Government shall be responsible for GB tests. Sample shall be forwarded to the laboratory designated by the contracting officer (see 6.1).

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4.1.3 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Preproduction lot. Prior to the start of regular production and following any change in regular production methods, materials or equipment, a preproduction lot of 144 canisters shall be produced using the same methods, materials and equipment as will be employed during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination and non-destructive tests. Each canister from preproduction lot shall be inspected in accordance with its applicable drawing and specification requirements.

4.3.2.2 For destructive tests. Representative canisters shall be taken from each preproduction lot and tested in accordance with table II. Eighteen (18) representative canisters shall be taken from each preproduction lot and forwarded to a Government test agency, as specified in the contract for GB life tests as specified in 4.4.4.1.

TABLE II. Preproduction testing

Test	Before rough handling	After rough handling
Volume of filling and charcoal moisture content	15	
DOP Penetration	All	20
CK gas life	18	
DMMP life	9	9
GB gas life	18	

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4.3.3 Acceptance/rejection criteria. If any one canister fails to meet 4.3.2.1 and 4.3.2.2 or if $x - 1.8s$ for each of the three gases CK, GB, and DMMP is less than the requirements of paragraph 3.9, the preproduction lot shall be rejected, where x is the measured mean gas life and "s" is the standard deviation as calculated in paragraph 6.5. The cause of failure shall be determined, corrective action instituted, and another preproduction lot produced and subjected to the same examination and testing as the initial preproduction lot. No regular production lot shall be accepted prior to approval of a preproduction lot. The contractor shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the canisters produced by one manufacturer, at one plant, from the same materials under essentially the same manufacturing conditions, and during one week production, except that no more than one lot of impregnated charcoal or one lot of filter material may be represented in one lot of canisters.

4.4.2 Sampling.

4.4.2.1 For examination and non-destructive tests. Sampling shall be conducted in accordance with MIL-STD-105. All unit packing canisters (5.1) shall be leak tested.

4.4.2.2 For destructive tests. Sampling for tests shall be in accordance with table III.

TABLE III. Regular production testing

Test	Before rough handling	After rough handling
Volume of filling and charcoal moisture content	8	
DOP penetration	All	16
CK gas life	8	
DMMP	8	8

4.4.3 Inspection procedure.

4.4.3.1 For examination and non-destructive tests. Sample canisters shall be examined in accordance with the classification of defects and MIL-STD-105. All unit packing canisters (5.1) shall be leak tested as specified in 4.4.4.2.

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4.4.3.2 For destructive tests. Canisters shall be taken from each lot and subjected to tests in the quantities indicated in table II.

4.4.3.3 For critical defects. Each canister in the lot shall be tested for critical characteristics listed in the classification of defects (4.4.3.4). Any canister failing to meet the specified test requirements shall cause the lot to be rejected.

4.4.3.4 Classification of defects.

(a) Canister, chemical-biological M10A1 Assembly (Dwg E5-3-547).

Categories	Defects	Acceptance standards
Critical:		
1	Canister leak	4.4.4.1
2	Canister resistance	4.4.4.1
3	DOP penetration before rough handling	4.4.4.1
Major: AQL 0.25 percent		
101	Valve leakage	4.4.4.1
102	Valve resistance	4.4.4.1
AQL 1.0 percent		
103	Component missing, incorrect, or incorrectly assembled	
104	Solder, paint or varnish missing, incorrect or inadequate	
105	Weight marking incorrect (canister weight shall be within 6 grams of the stamped weight	
Minor: AQL 2.5 percent		
201	Marking missing or illegible	
202	Contamination	

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(b) Purifier, air assembly (Dwg D5-3-549).

Categories	Defects
Major:	AQL 1.0 percent
101	Component missing, incorrect or incorrectly assembled
102	Damage
Minor:	AQL 2.5 percent
201	Contamination

(c) Container, chemical assembly (Dwg D5-3-548).

Categories	Defects
Critical:	
1	Chemical container top leakage
Major:	AQL 1.0 percent
101	Component missing, incorrect, or incorrectly assembled
102	Damage
Minor:	AQL 2.5 percent
201	Contamination

(d) Interior packaging (filled and sealed) assembly - (Dwg C5-12-434).

Categories	Defects	Acceptance standards
Major:	AQL 0.25 percent defective	
101	Container not correct	
102	Container not supplied with tear strip	
103	Corrosion resistant key not furnished and/or not attached to end of each can lid as specified	
	AQL 2.5 percent defective	
104	Packing incorrect	
105	Marking missing, incorrect or illegible	

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4.4.4 Tests.

4.4.4.1 Tests shall be as outlined in table IV.

TABLE IV. Test methods

Requirement	Method of test
Volume of filling (3.4)	Meter, Chemical Container, Volume Q134 (C38-5-296)
Charcoal moisture content (3.5)	Moisture Teller
Canister Resistance (3.7)	Indicator, Resistance, Canister, Air Purifier and Chemical Container, Q112 (B38-6-351-1) (a)
DOP Aerosol Penetration (3.10)	Penetrometer, Filter Tester, DOP, Q107 (B136-42-102-11)
Gas Life (CK) (3.9)	All Purpose Apparatus, Gas Life Testing, Q95 (B136-41-600-11) (b)
Gas Life DMMP and GB (3.9)	Test Apparatus Agent Gas Sorption Q110 (B136-41-850-3) (b)
Rough Handling (3.11)	Machine, Rough Handling, Q113 (C26-19-51) (c)
Chemical Container Leakage (3.3)	(d)
Canister Leakage (3.8)	(e)
Valve leakage (3.6.1)	Tester, Air Leakage, Flow Type, Q111 (B136-14-170-5)
Valve resistance (3.6.2)	Resistance Indicator, Inlet and Outlet Valve, Q212 (B136-44-1-5)

(a) Testing shall be conducted as follows: Draw air through the influent opening and out through the effluent opening at a rate equivalent to 85 liters per minute at 25°C and 760 millimeters of mercury (see 6.3). Do not continue the flow of air through the canister any longer than necessary to take resistance reading.

(b) The conditions for gas life testing shall be as follows:

Gas	Concentration	Temperature	Relative Humidity	Rate of Airflow
GB	4.0 ± 0.4 mg/liter	26 ± 6°C	15%	32 liters/min[1]
CK	4.0 ± 0.4 mg/liter	26 ± 6°C	80%	50 liters/min[2]
DMMP	3.0 ± 0.2 mg/liter	26 ± 6°C	15%	50 liters/min[1]

[1]Constant flow
[2]Intermittent flow

The gas life shall be reported to the nearest 0.1 minute.

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TABLE IV. Test methods (cont'd)

(c) The rough handling machine shall not be operated with more or less than four canisters in hermetically sealed container (C5-12-434) for a minimum period of 15 minutes.

(d) Prior to assembly, the canister joint between the chemical container top and nozzle shall be tested for leaks at air pressure differential of 5 psi for 10 seconds.

(e) After the canister is assembled, the top and bottom circumferential seams, the side seam and the joint between the nozzle and canister top shall be tested for leakage at air pressure differential of 5 psi for 10 seconds.

4.4.4.2 Packing container leakage. The leakage characteristics of the sealed container shall be determined by subjecting the samples taken in accordance with 4.4.2.1 to the leakage test by either the hot water or vacuum chamber technique of Federal Test Method Standard 101, Method 5009.

4.4.5 Acceptance/rejection criteria. If any canister fails volume of filling, charcoal moisture content, or the DOP penetration before rough handling test, the lot represented shall be rejected. If more than two of the sixteen canisters subjected to DOP penetration after rough handling fail that test, the lot represented shall be rejected. If $\bar{X} - 2.219s$ for both CK and DMMP is less than the requirements of paragraph 3.9, the production lot represented shall be rejected, where $\bar{X}$ is the measured mean gas life and "s" is the standard deviation as calculated in paragraph 6.5.

5. PACKAGING

5.1 Preservation, unit packing, and marking (level A, see 6.1). The filter canister shall be packaged and marked in accordance with C5-12-434.

5.1.1 Package test. The filter canister, unit packed as specified in 5.1, shall show no sign of leakage as evidenced by release of a continuous stream of bubbles which appear at any surface as specified in 4.4.4.2. The container that fails leakage test shall be repacked and retested.

5.2 Packing. Packing shall be level A, B or commercial, as specified (see 6.1).

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5.2.1 Level A. Eighteen (18) canisters, unit packed as specified in 5.1, shall be packed in a nailed wood box conforming to class 2, style 4, grade B, type 3 load of PPP-B-621. The box shall have inside dimensions of 22-3/4 by 11-1/2 by 7-1/2 inches, length, width and depth, respectively. The packaged canisters shall be separated by interlocking partitions fabricated of fiberboard conforming to type CF, class-weather resistant, variety SW, grade W5c of PPP-F-320. Separator pads of fiberboard (PPP-F-320) shall be placed between the packaged canisters and the sides, ends, top and bottom of the box. Additional pads shall be used as required to assure a tight pack. The box shall be closed and strapped in accordance with the appendix to PPP-B-621, using strapping and seals conforming to QQ-S-781 (strapping - type I (regular duty), finish B, grade 2 and seals - type D, styles I or IV, finish B, grade 2).

5.2.2 Level B. Eighteen (18) canisters, packaged as specified in 5.1, shall be packed in a fiberboard box conforming to variety SW, grade W5c, style RSC, type 3 load of PPP-B-636, having inside dimensions of 22-7/8 by 11-3/8 by 7-1/2 inches (length, width and depth, respectively). The packaged canisters shall be separated with interlocking partitions and filler pads as specified in 5.2.1. The box shall be closed in accordance with method V closure requirements and reinforced using nonmetallic or tape banding in accordance with reinforcing requirements in the appendix to PPP-B-636.

5.2.3 Commercial packing. Canisters, packaged as specified in 5.1, shall be packed for shipment in accordance with MIL-STD-1188. Unless otherwise specified, the quantity of packaged canisters in each commercial shipping container shall be eighteen (18).

5.3 Marking. In addition to any special marking required by the contract or order, unit packs (5.1) shall be marked in accordance with Drawing C5-12-434, and shipping containers (5.2.1, 5.2.2 and 5.2.3) shall be marked in accordance with MIL-STD-129.

5.4 Repair part. When this item is procured for storage and issue as a repair part, preservation, packaging, packing and marking shall be as specified on the packaging data sheet which is identified by the National Stock Number for this item.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of packing required (see 5.2).
(c) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.

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(2) Name and address of test facility and shipping instruct when testing is performed by the Government.

(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.2 Sampling and testing. When the contractor consistently produces high quality material and operates under a system of quality control acceptable to the Government, the Government, at its discretion, may modify in whole or in part, the sampling and testing procedures specified herein. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

6.3 Canister resistance. The resistance of the air purifier without filter should not exceed 16 mm of water, and with filter is should not exceed 71 mm of water in order to meet the final resistance requirement.

6.4 Canister closing. The assembly line should be so arranged that the canisters, after filling, are exposed to the atmosphere for the shortest possible time. Immediately after filling and closing, the canisters shall be sealed by suitable closures, placed in air tight containers such as charcoal drums, or other adequate containers, to prevent undue moisture absorption, pending placing the canisters in the appropriate packaging container. These precautions are necessary because charcoal can absorb sufficient moisture during the manufacturing operations to cause the canisters to be rejected.

6.5 Standard deviation. To find the standard deviation (S-) of a group of results:

Step 1. Square the sum of the individual observations $(\Sigma X)^2$.

Step 2. Multiply the sum of the squares of the individual observations by the number of observations.

Step 3. Subtract step 1 from step 2 and divide by the number of observations multiplied by the number of observations minus one N(N-1).

Step 4. Extract the square root of step 3. The process may be indicated thus:

$$s = \sqrt{\frac{N \Sigma x^2 - (\Sigma X)^2}{N(N-1)}}$$

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6.6 Air flow resistance. Atmospheric conditions often change rapidly and necessitate recalibration of the airflow resistance measuring device several times a day to insure correct readings. However, frequent recalibration can generally be avoided by use of a standard resistance for setting the flow manometer at the 85-liter point. If a standard resistance is used it must provide true laminar flow, be properly used to prevent turbulence, and the indicated resistance value checked frequently on a properly calibrated flowmeter.

6.7 Dimethylmethylphosphonate (DMMP) test reagent. DMMP may be purchased from Mobil Chemical Co., Industrial Chemicals Division, 801 E. Main Street, P.O. Box 1136, Richmond, VA 23208; or Virginia Chemicals Inc., 3340 W. Norfolk Road, Portsmouth, VA 23703.

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